

Deterministic Optimization

Large-Scale Optimization

Dantzig-Wolfe
Decomposition

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Exploiting Special Structures of
Large-Scale Optimization

Large-Scale Optimization

Dantzig-Wolfe

Learning Objectives for this module

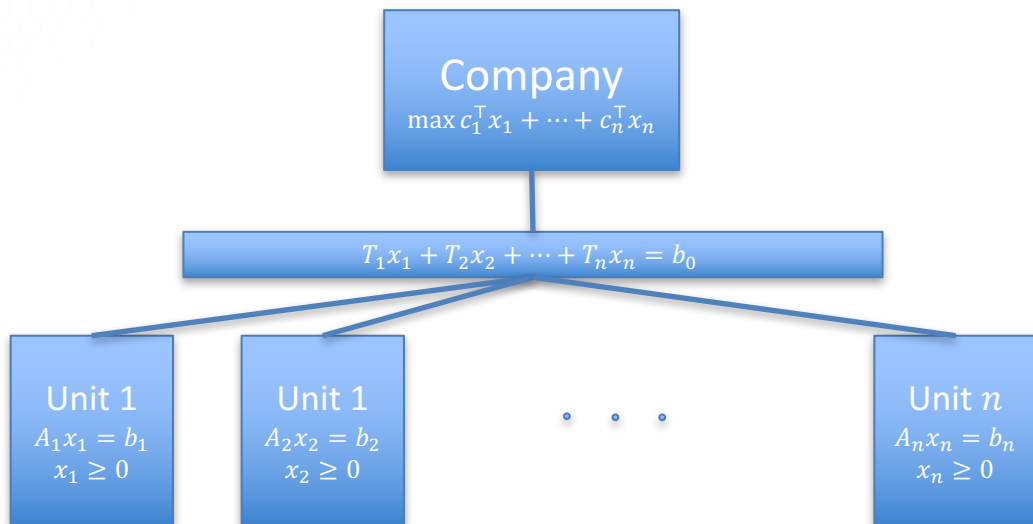
- Discover special structures of large-scale optimization problems
- Design a new decomposition algorithm that exploits special structures of the optimization problem

Central Company & Sub-Units

- Consider a large company that comprises n subsidiary units.
- Each subsidiary unit i has its own objective of their profit by planning its own production subject to its own constraints
 - Decision: x_i (e.g. how much to produce, store, sell, ...)
 - Local constraints: $A_i x_i = b_i, x_i \geq 0$ (e.g. satisfy demand, etc)
- All units share a common resource and there is a budget on the common resource (e.g. total monetary budget, total machine time, etc)
- The company want to maximize the total profit (sum of the sub-units' profits), respecting their local constraints, and the total budget constraint

Overall Problem

- The situation can be modeled as the following graph



$$\begin{aligned} \max \quad & c_1^\top x_1 + \dots + c_n^\top x_n \\ \text{s.t.} \quad & T_1 x_1 + \dots + T_n x_n = b_0 \\ & A_1 x_1 = b_1, \quad x_1 \geq 0 \\ & \dots \\ & A_n x_n = b_n, \quad x_n \geq 0 \end{aligned}$$

Special Structure of Constraints

- The constraints can be written in a block matrix form:

$$\begin{bmatrix} \boxed{T_1} & \boxed{T_2} & \cdots & \boxed{T_n} \\ \boxed{A_1} & & & \\ & \boxed{A_2} & & \\ & & \ddots & \\ & & & \boxed{A_n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

- This structure is called the **block angular** structure
- This can be a large-scale optimization problem with big n

Exploiting Special Structure

- The constraint $T_1x_1 + T_2x_2 + \dots + T_nx_n = b_0$ is called a **coupling** constraint
 - Because it couples all the sub-units together
 - Such a constraint is also called a **complicating** constraint
- If we do not have the coupling constraint, then all sub-unit constraints are **separated**
- If we can get rid off the complicating constraint, then the whole problem become decoupled into n separated, smaller problems

Distributed Optimization & Coordination

- An important algorithmic idea is **Coordination & Distributed Optimization**
 - Somehow get rid off the coupling constraint
 - Let each sub-unit solve its own problem in a distributed fashion
 - Coordinate sub-units' decisions so that the coupling constraint is accounted for

Summary

- We learned:
 - An important structure in large-scale optimization: Block angular structure
 - An outline of the coordinate & distributed optimization algorithm